

What Makes a Baseball Player Swing?

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Abstract

When a hitter steps into the batters box he knows once the pitch is thrown he is about to be faced with a binary decision. The hitter has a very small amount of time to make his decision. This paper analyzes what factors influence the hitter to make one choice or the other. In addition, which factors are valued more in the decision making process. This was solved by applying a drift-diffusion-style model to just under 50,000 four-seam pitches across the beginning of the 2023 season to the end of the 2025 season. The findings of the model show that hitters value both visual and contextual information. Specifically, pitches outside the zone and large pitch counts are seen as the most likely reasons hitters decide to swing or take.

Keywords: Binary Decision Making; Baseball; Evidence Accumulation Model; Drift Diffusion Model

Introduction

Every human on a day to day basis face several different binary decisions. Fight or flight? Chinese or Mexican? Each binary decision is made with a limited amount of time, hence why they are stressful. Thus, humans rely on the perceived information to make the correct choice. In this specified case of a baseball pitch the choice is swing or take.

Major League Baseball (MLB) first started tracking rudimentary pitch level data in 2008 (Major League Baseball, 2025a). Nearly each year the MLB is posting new variables and data to be able to tackle new problems. In fact, the newest update just released in 2024 (Powers & Yurko, 2025). Although it is possible to have analyzed this question before, the result was not complete. Now, with the new data it provides a clearer more whole picture. In addition to data constraints, the MLB has evolved in the last few decades. Across these decades there have been several new rule changes and additions. For instance, just as recently as 2023 there were three major rule changes including the pitch clock, ban on the shift, and larger bases (Castrovince, 2023). Thus, any research completed on this topic even only a decade ago is already outdated.

This paper will analyze the new components available with technology and MLB advancement that hitters utilize in their decision to swing or not swing through a Deadline-constrained evidence accumulation model (a drift-diffusion-style model). In this style of model it explains across the different kinds of information a hitter is presented with what information makes the hitter more likely to swing and what information makes the hitter more likely to take.

From this sort of model it explains what is the most important information to a hitter. This is on levels of not only magnitude but direction. This insight can help identify possible bias in the hitter's approach. By highlighting what information the hitter values this can lead to further exploration as well as development of the hitter. In the following sections this paper

will clarify the process of the research through the background, approach, methods, results, and discussion.

Background

On each pitch hitters have between 125 to 225 milliseconds to decide on whether to swing or take the pitch (Rodriguez, 2025). This clearly makes the deadline for this binary decision very short making it difficult to make the right choice. However, the choice made from the decision can have lasting ramifications to the outcome of not only the hitter's at-bat but the game as a whole. To clarify, on each pitch a hitter faces it can result in one of many outcomes dependent on their decision. If the hitter chooses to take the pitch the results are either a ball or taken strike. If the hitter chooses to swing the results include a swing and miss for a strike or a ball-in-play (BIP). If the hitter combines for three swings and misses or taken strikes his at-bat concludes in a strikeout. Furthermore, a BIP can result in a hit or an out. However, a hit is not the only beneficial outcome for a hitter. If the hitter is able to take four balls before he earns three strikes he earns a walk and is now on first base. Now, BIP hit outcomes such as singles, doubles, triples, and home runs produce higher run values in comparison to a walk, but to create those outcomes you must swing (Pinheiro & Szymanski, 2022). While, on the other hand, besides a select few players, swinging the bat actually results in negative run value (Haque, 2022). This further stresses how crucial it is to make a good decision when deciding whether or not to swing.

When hitters are faced with the critical decision they utilize every piece of information available. This information is a mixture of different categorical inputs the brain processes.

Contextual Input

The first input the hitter takes into consideration before determining whether to swing or take the pitch is the context of the current game state. This means, during every pitch throughout a baseball game there is a one of one specific scenario. This scenario can alter due to the inning (point we are at in the game), the number of outs when they are up to bat, the current count of the at-bat, the score of the game, and many more. Several pieces of contextual information, such as the inning, is static throughout the entire at-bat. While, on the other hand, many different pieces of input can change pitch to pitch. Ultimately, all of these different pieces of information are known to the hitter before he has to make the decision on the pitch. Many, if not all, of these pieces of contextual information can be sizable in influencing the hitter's decision.

Adding further complications, not only is the hitter conscious of these streams of information, the pitcher is also knowledgeable of this information. This further escalates the

situation as pitchers can exploit hitters if they rely too much on the contextual information. For instance, a hitter may be in a favorable three balls and one strike count, meaning one more ball and the hitter is granted a walk. This may lead the hitter into thinking the pitcher must throw a strike; otherwise, the hitter will be granted first base. Exploiting this perceived notion, the pitcher purposefully throws a ball fooling the hitter to swing at a bad pitch. These interactions between pitchers and hitters constantly occur in the MLB only complicating the decision further. The hitter, simply, cannot rely on a single piece of contextual information when making his decision to swing.

Visual Input

The second input the hitter takes under review when determining whether to swing or take the pitch is the location of the ball. Unlike contextual information, such as the count, this input is continuously changing until the deadline has expired. To clarify, in the early stages of contemplation a pitch may be viewed as a good pitch to swing on to try and produce a quality BIP outcome. However, as time passes the pitch may look not as worthy of a swing. This continuous change in potential outcome of a pitch drastically entangles the decision to swing.

Furthermore, it is important to point out, the location of the pitch is the last piece of information the hitter views before the deadline. Recency bias is a well studied phenomenon with several studies suggest that individuals over value information seen recently rather than earlier (Yoo, Bahg, Turner, et al., 2025). Thus, to the hitter, this same pattern applies with the location of the ball remembered and seen as more valuable in comparison to the earlier contextual information.

Approach

As mentioned previously, to discover the reasoning behind what makes a player swing we utilized a Deadline-constrained evidence accumulation model (a drift-diffusion-style model). In this model as the hitter walks up to the plate they start at a point in between two boundaries. The upper boundary represents the decision to swing and the lower boundary represents the decision to take. The starting point between the two boundaries is determined by the prior game-context information. To clarify, if the contextual information before the at-bat begins gives the hitter a reason to swing the hitter will drift towards the upper boundary before even seeing the pitch. This same process occurs pitch after pitch with new updated information, both contextual and visual, pushing the hitter to one boundary or the other before the deadline.

This model is incredibly useful in this swing-take scenario due to three different reasons. The first is the swing-take decision is a binary decision. Thus, there is no third or fourth boundary complicating the problem exponentially as the hitter is stuck between multiple choices. Second, given that there is a time constraint on the decision the hitter must drift towards one boundary or the other. Simply, there is no indecision. If the hitter freezes then their decision is to take the pitch. Lastly, in order for the hitter to drift to one boundary or the other it relies

on multiple multi categorical inputs. For instance, if many of the contextual inputs are pointing to the decision to swing and yet the hitter's visual input at each step is determining the pitch is going to be in the dirt the hitter may reverse the way he was originally drifting. Ultimately, this model is a great fit for this sort of decision as both choices result in beneficial outcomes for the hitter when the right choice is selected.

Although the model is an excellent match for this binary decision, an additional central factor in this exploration is not only what inputs make the hitter drift towards one boundary or the other, but the magnitude of that drift. Not all information is created equal in the mind of the hitter. Much of the value assigned to certain categorical information occurs before the ball leaves the hand of the hitter. Are these various inputs valued in the appropriate manner to select the optimal choice, or is it wrapped in systematic bias negatively affecting the hitter's swing decision? Ultimately, the goal of this approach is to explore that question utilizing the deadline constrained evidence accumulating process.

Method

Data Collection & Processing

To explore the wide range of inputs that impact the hitter's decision to swing or take a pitch, pitch-level data was acquired from MLB Statcast via Baseball Savant (Major League Baseball, 2025b). The data was downloaded using the "pybaseball" Python library. The data set covers every pitch from the start of the 2023 regular season (March 20, 2023) to the end of the 2025 regular season (September 30, 2025). The reasoning behind the selection of this date range was the new rule changes put in place at the start of the 2023 season, mentioned previously.

Following the download procedure, there was a processing of relevant variables. Meaning, certain pitch-level data such as hit distance was removed from the data set as this was not in the scope of my research. The relevant data remaining was either contextual or visual information the hitter utilizes in their decision to swing.

Next, any pitch event that was not classified as a four-seam fastball was removed from the data. This is due to the characteristics of this pitch versus other pitches. To clarify, the trajectory of a four-seam fastball is nearly a straight line from the pitcher's hand to the catcher's glove. Thus, the final location of the pitch is very similar to where the hitter views the pitch. Due to there being no public data that can pinpoint the exact location of the ball during certain moments of a pitch this study assumes the final location of the ball is the visual input for the hitter. Furthermore, many other pitches have spin and break; meaning, the location of the ball mid-pitch is very different in comparison to its final location. Thus, for this study we controlled for only four-seam fastballs.

The next crucial step taken in the data processing operation was the addition of a binary swing variable. In the data from Statcast there was no feature to flag for swings versus takes thus one was created. This was through the description vari-

able of each pitch. When the pitch event was described as a "swinging_strike", "hit_into_play", "foul", and many more the swing variable was given a value of one. Otherwise, to represent balls, the swing variable was given a value of zero. This variable establishment was crucial to determine what inputs hitter's possessed and the decision they made.

A second variable was created that represented the time it took the ball to reach home plate. The equation for this calculation was:

$$t = \frac{D - e}{v_{\text{mph}} \cdot c}.$$

Where $D = 60.5$; Distance from the pitcher's mound to home plate(ft), $c \approx 1.4667$; Conversion factor from miles per hour (mph) to feet per second (ft/s), e is release extension (ft), and v_{mph} is pitch velocity (mph). The release extension and pitch velocity are both variables acquired from Statcast. This variable had a dual purpose: filtration and analysis. First, any data point (pitch) that had missing information as well as an unrealistic time was removed from the data set. Second, it allows for investigation of the visual input over time. The variable served as a simplification of reaction time. Meaning, as time got closer to the deadline we could examine the possible change in value in regards to the ball location.

Verification of the Drift-Diffusion-Style Model

Prior to the insights provided by the final model critical steps were taken to verify the data was appropriate as well as demonstrates for the need of a complex cognitive model.

Baseline Logistic Regression This rudimentary model was designed to validate the data. There needed to be evidence of relationships between specific contextual and visual inputs with the hitter's choice. For instance, for more balls pitched into the strike zone hitter's should be swinging more. These sort of simple relationships to between features was crucial to verify the accuracy of the data before proceeding with the next step.

Location, Reaction Time, and their Relationship The purpose of this step was to show the development of visual inputs over time. Prior to the pitch the hitter already has all of the contextual information. Meaning, the hitter has their starting point between both boundaries. As the pitch is thrown, it first evaluates across location to see if location can predict swings. Location is distinguished between three zones: heart (middle of strike zone), shadow (edges of strike zone), and chase (outside the strike zone). Next, it evaluates the role the time it took for the ball to reach home plate (reaction time simplification variable) in the decision making process. Lastly, it evaluates the relationship between the two variables. For instance, it determines if a hitter has less time to make a decision and has yet to reach a boundary does the decision become harder. In addition, it is important to note, this model holds constant contextual inputs.

Implementation of Drift-Diffusion-Style Model

Following the previous steps it became evident that a more complex model was needed. The Deadline-constrained evidence accumulation model (a drift-diffusion-style model) utilized 28 different parameters grouped into two categories. Group one contains the core parameters for a drift-diffusion-style model: boundary separation (a) or distance between the upper and lower boundaries and time before evidence accumulation begins (t_0). a was originally set at an arbitrary moderate value of 2.0. a was then changed over time to 0.118 using a maximum likelihood estimation (MLE), which is shown below. The reason the valued narrow was due to the super quick deadline. t_0 was set to 0.1 or 100 milliseconds. This was due to, on average, it takes 100 milliseconds for the hitter to process the baseball (Rodriguez, 2025). Now with t_0 it is possible to calculate the time available until the deadline:

$$T_{\text{decision}} = \text{time_to_plate_sec} - t_0$$

Group two contains the remaining 25 parameters that are the drift rate coefficients. β_0 is the intercept or baseline drift. The base state selected was 0-ball 0-strike count, 0 outs, no runners on, and a pitch in the strike zone, with the average reaction time and release speed. This base state was chosen as it is the first at bat of the inning. This is the most simple game context. The remaining drift rate parameters are all based off this base state. To solve for drift rate (v) it is a linear combination of features seen as:

$$\begin{aligned} v = & \beta_0 + \beta_S \cdot \text{shadow} + \beta_C \cdot \text{chase} \\ & + \beta_{RT} \cdot \text{time_to_plate_sec_std} \\ & + \beta_{RTS} \cdot (\text{time_to_plate_sec_std} \times \text{shadow}) \\ & + \sum (\beta_{\text{Count}} \cdot \text{count_dummies}) \\ & + \sum (\beta_{\text{Outs}} \cdot \text{outs_dummies}) \\ & + \sum (\beta_{\text{Runner}} \cdot \text{runner_dummies}) \\ & + \beta_{\text{Speed}} \cdot \text{release_speed_std}. \end{aligned}$$

Where β_0 denotes the intercept (baseline drift), β_S captures the effect of shadow location, β_C captures the effect of chase location, β_{RT} represents the main effect of reaction time, and β_{RTS} represents the interaction between reaction time and shadow location. The model further includes eleven count-state coefficients, two outs coefficients, seven runner-state coefficients, and one coefficient for standardized release speed.

Now, with equations above it is possible to solve for the probability of the hitter choosing to swing before time T. This is done in two parts. First, utilizing the steady-state probability (infinite time limit): For zero drift ($v \approx 0$),

$$P_{\text{upper}} = \frac{z_{\text{abs}}}{a}.$$

For nonzero drift,

$$P_{\text{upper}} = \frac{1 - \exp\left(-\frac{2vz_{\text{abs}}}{s^2}\right)}{1 - \exp\left(-\frac{2va}{s^2}\right)}.$$

Where $z_{\text{abs}} = z \cdot a$. This is the starting point in absolute units, z is the starting point expressed as a fraction of the boundary separation (fixed at $z = 0.5$; midpoint between boundary edges), a is the boundary separation, s is the noise parameter (fixed at $s = 1.0$; standard for drift-diffusion-style models), and v is the drift rate.

Then, add in the time dependent correction:

$$P = P_{\text{upper}} + \sum_{n=1}^N \text{term}_n.$$

Where each term is:

For zero drift ($v \approx 0$),

$$\text{term}_n = \frac{2}{\pi n} \exp\left(-\frac{\lambda_n^2 T_{\text{decision}}}{2}\right) \sin(\pi n z).$$

For nonzero drift,

$$\text{term}_n = \frac{2\pi n}{a^2} \exp\left(-\frac{\lambda_n^2 T_{\text{decision}}}{2}\right) \sin(\pi n z) \times \frac{\exp(v(a - z_{\text{abs}})) - (-1)^n \exp(-v z_{\text{abs}})}{\exp(va) - \exp(-va)}.$$

And:

$$\lambda_n = \frac{\pi n s}{a}.$$

The series is usually truncated after ten to 15 terms, once additional terms become negligible.

This probability P computed above is the probability that the hitter chooses to swing given all the different categorical inputs. This probability is then compared to the actual decision made by the hitter. The MLE then learns what inputs influence a hitter to swing and what inputs a hitter to take the pitch. In addition, which inputs are valued most by the hitter. In this paper the log-likelihood was preferred due to the stability with P 's that were negative:

$$\ell(\theta) = \sum_{i=1}^N [y_i \log(P_i) + (1 - y_i) \log(1 - P_i)].$$

Where y is the actual decision made by the hitter. This process was repeated for every pitch i over the nearly 50,000 pitch data set.

Results

As shown in Table 1, location plays a significant role in altering the drift rate. Pitches viewed far outside the strike zone (in chase zone) is greatest indicator for a hitter to take a pitch. Furthermore, the length of time it takes for the pitch to get to the plate also plays a significant role in changing the drift rate.

As shown in Table 2, the full count of three balls and two strikes show the highest likelihood of the hitter choosing to swing in comparison to the base count of zero balls and zero strikes. In fact, nearly every count besides the three ball zero strike count demonstrates a greater likelihood of swinging the bat.

Table 1: Drift Rate Coefficients

Parameter	Symbol	Coefficient
Intercept (baseline)	β_0	-1.1258
Shadow zone	β_S	0.3195
Chase zone	β_C	-7.5020
Time to plate	β_{RT}	-1.1303
Time $\times$ Shadow	β_{RTS}	0.2027

Note. Time-to-plate coefficient is standardized (mean = 0.4125 s, SD = 0.0288 s).

Table 2: Count State Coefficients (relative to 0-0 baseline)

Count	Coeff.	Count	Coeff.	Count	Coeff.
0-0	0.000	1-0	2.864	2-0	2.692
0-1	3.323	1-1	4.326	2-1	5.434
0-2	3.575	1-2	4.900	2-2	6.876
3-0	-0.855	3-1	2.587	3-2	6.929

Table 3: Runner State Coefficients (relative to empty bases baseline)

Runner State	Description	Coefficient
Empty (baseline)	No runners on base	0.000000
1	Runner on first	0.000000
2	Runner on second	0.000000
3	Runner on third	0.000000
12	Runners on 1st & 2nd	0.000000
13	Runners on 1st & 3rd	0.000000
23	Runners on 2nd & 3rd	0.000000
123	Bases loaded	-0.431229

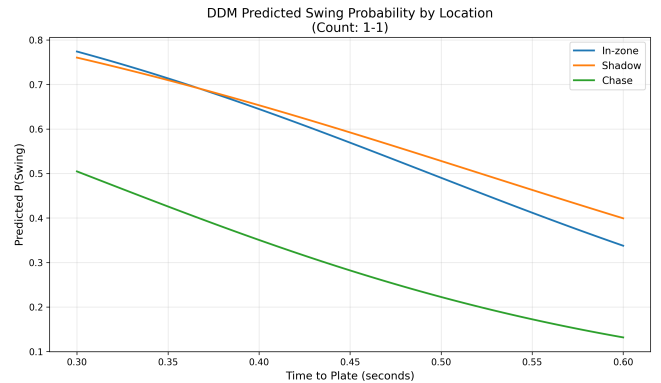


Figure 1: DDM-predicted swing probability as a function of time to plate for three pitch location categories (in-zone, shadow, and chase) under a one ball and one strike count.

As shown in Table 3, for the most part there is no difference in when a hitter chooses to swing across different base runner states. The exemption is when the bases are loaded hitters are slightly less likely to swing.

Table 4: Model Fit Statistics

Statistic	Value
Number of trials	49,565
Number of parameters	27
Log-likelihood	-31,338.78

In Figure 1, the probability of a hitter swinging at a pitch is highly dependent on the location of the pitch, as found earlier. With the following order: in-zone, shadow, and then chase. The probability of the hitter choosing to swing decreases as the time it takes for the pitch to get to home plate increases. This result is constant across all different plate location zones.

As seen in Table 4 is the model fit statistics. The Log-likelihood is very large. However, this is expected due to the complexity of the model as well as the number of trials. When the Log-likelihood is evaluated on a trial by trial basis the result is appropriate.

Discussion

Ultimately, the goal of this paper was to determine what hitters found important when making the decision to swing or take. With this knowledge it could set in motion future research and development for the player. For instance, if a hitter knew he held bias towards certain counts he could alter his at-bat approach to further improve as a player.

In regards to the results, it was interesting to see how valuable both the visual and contextual information was to a hitter. In Table 2 it is important to point out that with the base case being the zero ball zero strike count, regardless of the result of the next pitch, on the future pitch at either zero balls and one strike or one ball and zero strikes a hitter was more likely to swing. There is clearly a pattern as the number of total pitches increase in an at-bat the more likely a hitter was to swing besides the outlier mentioned in the results.

The results in Table 3 were surprising. The scenario of bases loaded leading a hitter to be less likely to swing makes sense as a walk now results in a run. However, it was shocking to see scenarios such as a runner only on third did not make the hitter more likely to swing. As the increased run value of a hitter earning a walk in that position is not significant, while any kind of hit is greatly beneficial.

It is also important to point out that the variables or inputs that were not included in the results such as outs had little to no effect on the hitter's probability of swinging, which makes sense.

There are two main limitations of this study. The first main limitation is the t_0 value. As described earlier this is the time until information accumulation began. The value represents when the visual information can begin accumulation once the pitch has been thrown. Although, this is accurate, the contextual information is known prior to the event even starting. This nuance could not be included into the model. The second limitation is there is no by player breakdown. Baseball players

are all very different their starting point (z) is not necessarily in the middle of both boundaries. Some players simply do not need as much evidence to choose to swing; while others, are very disciplined and need more evidence. Now, this could be a possible bias in their game they could develop, but in this model $z = 0.5$ simplifies the already complex model.

The alteration of player specification is the direction of future research in this area. Research done by an MLB team on their players or the teams they are competing against could be exponentially beneficial. For instance, as a pitcher, knowing the opposing hitter is less likely to swing in a certain at-bat situation could be exploited.

In conclusion, a hitter deciding whether to swing or take a pitch is like any binary decision. He utilizes information across many categorical areas to make the decision in a time constraint. The ability to make the right decision could lead the hitter to a World Series.

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